Assignment : Handling Missing Data

SECTION A – THEORETICAL QUESTIONS

Q1. What are the most common reasons for missing data in ETL pipelines?

Answer:
The most common reasons for missing data in ETL pipelines include data entry errors, incomplete forms, system or extraction failures, issues during data merging or joining, unavailability of data at the source, customer unwillingness to share information, and data loss during transfer between systems.

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

Answer:
Blindly deleting rows with missing values is a bad practice because it can lead to loss of important data, reduce the dataset size significantly, introduce bias if the missing data is not random, and negatively affect the accuracy of analysis and decision-making.

Q3. Explain the difference between:

Listwise deletion

Column deletion

Answer:
Listwise deletion refers to removing entire rows from the dataset if they contain any missing values in selected variables. It is appropriate when only a small number of rows have missing data and the dataset remains sufficiently large after removal.

Column deletion involves removing an entire column when a large portion of its data is missing or when the variable is not important. It is appropriate when a column has excessive missing values and cannot be reliably imputed.

Also mention one scenario where each is appropriate.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

Answer:
Median imputation is preferred for skewed data because it is less affected by extreme values or outliers. In datasets like income, where a few high values can distort the average, the median provides a more accurate representation of the central tendency.

Q5. What is forward fill and in what type of dataset is it most useful?

Answer:
Forward fill is a method of imputing missing values by replacing them with the last observed valid value. It is most useful in time-series or sequential datasets, such as stock prices, daily sales, or sensor data, where values tend to follow a logical sequence over time.

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

Answer:
Flagging missing values before imputation is important because it preserves information about which values were originally missing. This helps in identifying patterns of missingness,

allows the missingness to be used as a feature in analysis, and prevents loss of potentially valuable insights after imputation.

Q7. Consider a scenario where income is missing for many customers.

How can this missingness itself provide business insights?

Answer:
Missing income data can provide insights into customer behavior and data quality. It may indicate privacy concerns, reluctance to disclose financial information, or issues with data collection methods. It can also highlight specific customer segments or groups where missing data is more common, helping businesses improve their data collection strategies and better understand their customers.

SECTION B – PRACTICAL QUESTIONS

Customer_ID	Name	City	Monthly_Sales	Income	Region
101	Rahul Mehta	Mumbai	12000	65000	West
102	Anjali Rao	Bengaluru	NaN	NaN	South
103	Suresh Iyer	Chennai	15000	72000	South
104	Neha Singh	Delhi	NaN	NaN	North
105	Amit Verma	Pune	18000	58000	NaN
106	Karan Shah	Ahmedabad	NaN	61000	West
107	Pooja Das	Kolkata	14000	NaN	East
108	Riya Kapoor	Jaipur	16000	69000	North

Use the given dataset for all questions.

Q8. Listwise Deletion

Remove all rows where Region is missing.

Tasks:

1. Identify affected rows
2. Show the dataset after deletion
3. Mention how many records were lost

Answer :

Records lost:

- Before = 8 rows
- After = 7 rows
- **Deleted = 1 row**

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Customer	ID	Name	City	Monthly_Sales	Income	Region
101		Rahul Mehta	Mumbai	12000	65000	west
102		Anjali Rao	Bengaluru	NaN	NaN	South
103		Suresh Iyer	Chennai	15000	72000	South
104		Neha Singh	Delhi	NaN	NaN	North
106		Karan Shah	Ahmedabad	NaN	61000	West
107		Pooja Das	Kolkata	14000	NaN	East
108		Riya Kapoor	Jaipur	16000	69000	North

Q9. Imputation

Handle missing values in Monthly_Sales using:

- Forward Fill

Tasks:

1. Apply forward fill
2. Show before vs after values
3. Explain why forward fill is suitable here

Answer :

After :

Customer	ID	Name	City	Monthly_Sales	Income	Region
101		Rahul Mehta	Mumbai	12000	65000	west
102		Anjali Rao	Bengaluru	12000	NaN	South
103		Suresh Iyer	Chennai	12000	72000	South
104		Neha Singh	Delhi	15000	NaN	North
106		Karan Shah	Ahmedabad	15000	61000	West
107		Pooja Das	Kolkata	14000	NaN	East
108		Riya Kapoor	Jaipur	16000	69000	North

Before :

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A	B	C	D	E	F	G
Customer	ID	Name	City	Monthly_Sales	Income	Region
101		Rahul Mehta	Mumbai	12000	65000	west
102		Anjali Rao	Bengaluru	NaN	NaN	South
103		Suresh Iyer	Chennai	15000	72000	South
104		Neha Singh	Delhi	NaN	NaN	North
106		Karan Shah	Ahmedabad	NaN	61000	West
107		Pooja Das	Kolkata	14000	NaN	East
108		Riya Kapoor	Jaipur	16000	69000	North

Forward fill is suitable here because the **Monthly_Sales data follows a sequential pattern**, where each record comes in order (similar to time-based or ordered customer data). It assumes that the **previous valid value is the best available estimate** for the missing entry.

Since sales values usually do not change drastically between consecutive records, using the last known value helps **maintain continuity and avoids unrealistic gaps** in the dataset. It is also a simple and efficient method when no additional information is available for more complex imputation.

Q10. Flagging Missing Data

Create a flag column for missing Income.

Tasks:

1. Create Income_Missing_Flag (0 = present, 1 = missing)
2. Show updated dataset

ID	Name	City	Monthly_Sales	Income	Region	Income_Missing_Flag
	Rahul Mehta	Mumbai	12000	65000	west	0
	Anjali Rao	Bengaluru	NaN		South	1
	Suresh Iyer	Chennai	15000	72000	South	0
	Neha Singh	Delhi	NaN		North	1
	Karan Shah	Ahmedabad	NaN	61000	West	0
	Pooja Das	Kolkata	14000		East	1
	Riya Kapoor	Jaipur	16000	69000	North	0

3. Count how many customers have missing income

F11 | =COUNTIF(H1:H8,1)

	B	C	D	E	F	G
1				how many customers have missing income	3	
2						